Nanyang Technological University
SPMS/Division of Mathematical Sciences

2023/2024 Semester 2 | **MH1201** | **Mid-term**

29 Feb 2024 | TIME ALLOWED: 60 MINUTES

Name:	**Matriculation Number**:

Question	1	2	3	Total
Marks				

INSTRUCTIONS TO CANDIDATES

1. This test paper contains **THREE (3)** questions and comprises **SEVEN (7)** printed pages, including the cover page.
2. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
3. You may write on the opposite page or any page if there is not enough space.
4. This is a **CLOSED BOOK** exam.
5. You may use calculators.

QUESTION 1. **(15 Marks)**

Let $\boldsymbol{T} : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ be the map defined by $\boldsymbol{T}(f(x)) = f(x^2)$.

(a) Show that $\boldsymbol{T}$ is a linear map.

(b) Let $\mathcal{B} = \{1, x, x^2\}$ be the standard basis for $\mathcal{P}_2(\mathbb{R})$ and $\mathcal{A} = \{1, x, x^2, x^3, x^4\}$ be the standard basis for $\mathcal{P}_4(\mathbb{R})$. Write down the matrix representation of $\boldsymbol{T}$ with respect to $\mathcal{B}$ and $\mathcal{A}$.

QUESTION 1. (CONTINUED)

(c) Determine a basis for either null($\boldsymbol{T}$) OR range($\boldsymbol{T}$).

QUESTION 1. (CONTINUED)

(d) Determine the dimensions of null($\boldsymbol{T}$) AND range($\boldsymbol{T}$).

(e) Determine whether the following properties of $\boldsymbol{T}$ is true. You are to explain your answer.

(i) Is $\boldsymbol{T}$ injective?

(ii) Is $\boldsymbol{T}$ surjective?

QUESTION 2. **(10 Marks)**

Recall that $\mathcal{M}_{24\times 24}(\mathbb{R})$ is the vector space comprising all (24×24)-matrices. Also, recall that $\mathcal{S}_{24\times 24}(\mathbb{R})$ is the vector space comprising all (24×24)-symmetric matrices.

(a) Let $\boldsymbol{E}_{1,1}$ be the matrix with one at the first row and the first column, and zero elsewhere. In other words, $\boldsymbol{E}_{1,1}$ has exactly one 1.

Consider the set $\mathcal{U} = \{\boldsymbol{A} + \boldsymbol{E}_{1,1} \in \mathcal{M}_{24\times 24}(\mathbb{R}) \ : \ \boldsymbol{A} \in \mathcal{S}_{24\times 24}(\mathbb{R})\,\}$. Is $\mathcal{U}$ a subspace of $\mathcal{M}_{24\times 24}(\mathbb{R})$?

QUESTION 2. (CONTINUED)

(b) Let $\boldsymbol{E}_{20,24}$ be the matrix with one at the 20-th row and the 24-th column, and zero elsewhere. In other words, $\boldsymbol{E}_{20,24}$ has exactly one 1.

Consider the set $\mathcal{V} = \{\boldsymbol{A} + \boldsymbol{E}_{20,24} \in \mathcal{M}_{24\times 24}(\mathbb{R}) \ : \ \boldsymbol{A} \in \mathcal{S}_{24\times 24}(\mathbb{R})\,\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{24\times 24}(\mathbb{R})$?

QUESTION 3. **(5 Marks)**

Let $[\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3]$ be a basis for the vector space $\mathcal{V}$. Show that $[\boldsymbol{v}_1 + \lambda\boldsymbol{v}_3, \boldsymbol{v}_2 + \mu\boldsymbol{v}_3, \boldsymbol{v}_3]$ is a basis for $\mathcal{V}$ for any real values λ and μ.